MUSABE SCHOOLS

Academic Performance Tracking and Learning Analytics System

Project Concept and System Vision

A data-driven academic platform for monitoring student progress, identifying learning gaps, and supporting timely academic intervention.

Prepared as an initial concept for stakeholder and developer review

1. Project Overview

The Musabe Schools Academic Performance Tracking and Learning Analytics System is a web-based educational technology platform designed to monitor, evaluate, and analyze students' academic progress across the school's educational levels.

The primary objective is to move beyond traditional examination results and provide detailed insights into students' academic performance, learning difficulties, progress over time, and areas requiring improvement.

The system will enable teachers and school management to identify students who are falling behind academically, determine subjects and topics in which students experience difficulties, and make informed decisions to improve teaching and learning outcomes.

The platform will initially be developed as a web application, with a mobile application potentially introduced later for viewing academic analytics, reports, and performance summaries.

The system is intended to support Musabe Schools' Nursery and Primary education, as well as Secondary education, including Form I to Form IV.

2. Problem Statement

Traditional academic assessment systems primarily focus on examination marks, grades, and overall student rankings. Although these results provide a general indication of academic performance, they do not adequately explain:

- Why a student is performing poorly.
- Which subjects require additional attention.
- Which specific topics or learning objectives the student has not mastered.
- Which examination questions students frequently answer incorrectly.
- Whether a student's academic performance is improving or declining over time.
- Which students require additional academic support.
- Whether an entire class is struggling with a particular topic.
- Which teaching areas may require revision or improvement.

Musabe Schools requires a system that transforms examination results into actionable academic intelligence.

3. Main Objectives

- Monitor academic performance at individual student, class, subject, and school levels.
- Track student performance across multiple examinations and academic periods.
- Identify students who are academically behind their expected performance.
- Analyze examination performance at question and topic levels.
- Identify subjects, topics, and learning objectives that require improvement.
- Provide teachers with evidence-based insights for academic interventions.
- Enable school management to monitor overall academic performance.
- Support data-driven academic planning and decision-making.
- Track academic improvement throughout the academic year.
- Support preparation for Tanzania's national examinations, particularly NECTA examinations.

4. Examination and Assessment Structure

A central concept of the system is the examination or assessment set. A **Set represents a complete examination**, similar to a school examination or NECTA examination. A set does not represent individual questions or topic-based exercises.

Examples of examination sets include:

- Monthly Examination 1
- Monthly Examination 2
- Mid-Term Examination
- Terminal Examination
- Annual Examination
- Mock Examination
- Pre-NECTA Examination

The system will analyze performance across these examinations to determine whether students are improving, maintaining their performance, or declining academically.

Examination Set	Assessment
Set 1	Monthly Test 1
Set 2	Mid-Term Examination
Set 3	Monthly Test 2
Set 4	Terminal Examination
Set 5	Annual Examination

5. NECTA-Oriented Examination Approach

The system will primarily focus on examinations designed according to the Tanzania National Examinations Council (NECTA) examination structure and standards. Teachers will be responsible for preparing and selecting examination questions.

The platform will not initially function as an automatic question-generation system.

Teachers will be able to:

- Create examinations.
- Add examination questions.
- Select questions from an available question bank.
- Use NECTA-style questions.
- Import questions where supported.
- Assign marks to individual questions.
- Define examination instructions and duration.
- Associate questions with subjects, topics, subtopics, and learning objectives.

The system should support examination structures appropriate for Standard 2, Standard 4, Standard 7, Form 2, and Form 4. Additional classes and educational levels may be supported as the system evolves.

6. Academic Data Structure

The system should organize academic information using the following hierarchy:

School → Academic Year → Term → Class → Subject → Examination → Questions → Student Results

Each examination question should optionally be associated with:

- Subject
- Topic
- Subtopic
- Learning objective or competency
- Question number
- Question type
- Difficulty level
- Maximum marks
- Marking scheme

This structure is essential because it allows the system to analyze examination performance beyond overall marks. For example, a Form 2 Mathematics examination may contain questions related to Algebra, Geometry, Statistics, Fractions, and Linear Equations. Although the examination is conducted as a single complete assessment, its individual questions can be mapped to their respective topics.

This allows the system to determine which topics students understand and which topics require additional instruction.

7. Student Performance Tracking

Every student should have an academic performance profile containing:

- Student identification information.
- School and class information.
- Academic year and term.
- Examination history.
- Subject performance.
- Examination marks and grades.
- Topic-level performance.
- Question-level performance.
- Academic progress over time.
- Identified learning difficulties.
- Recommended academic interventions.
- Previous and current performance comparisons.

The system should make it possible to view an individual student's academic history across different examination sets.

Examination	Mathematics
Monthly Test 1	42%

Examination	Mathematics
Mid-Term	51%
Monthly Test 2	63%
Terminal Examination	71%

The system should recognize improvement and identify students whose performance consistently declines or remains below the expected academic threshold.

8. Academic Analytics and Intelligence

This is the most important component of the system. The platform should provide several analytical perspectives.

8.1 Student-Level Analytics

- Students performing below the expected level.
- Students with declining academic performance.
- Students demonstrating consistent improvement.
- Subjects in which individual students perform poorly.
- Topics that individual students have not mastered.
- Students requiring remedial teaching or additional academic support.

8.2 Subject-Level Analytics

- Average marks and average percentage.
- Highest and lowest performance.
- Pass rate and grade distribution.
- Performance trends across examination sets.
- Topics with the lowest performance.
- Questions with the highest failure rates.

8.3 Topic-Level Analytics

The system should identify topics in which students experience difficulties, and determine whether the difficulty is limited to individual students or affects the entire class.

Topic	Class Average	Interpretation
Fractions	35%	Requires attention
Algebra	78%	Good understanding
Geometry	48%	Needs improvement
Statistics	82%	Strong performance

8.4 Question-Level Analytics

The system should analyze each examination question and display:

- Percentage of students answering correctly.
- Percentage of students answering incorrectly.

- Average marks obtained.
- Number of students who scored zero.
- Number of students who achieved full marks.
- Difficulty level based on student performance.

Question	Students Correct	Performance
Question 1	85%	Strong
Question 2	72%	Good
Question 3	31%	Weak
Question 4	24%	Critical

This information helps teachers identify questions or learning objectives that require additional explanation.

8.5 Class-Level Analytics

- Overall class performance.
- Class average across examinations.
- Subject-specific performance.
- Distribution of grades.
- Number of students below the academic target.
- Comparison between examination sets.
- Common learning difficulties.
- Students requiring academic intervention.

8.6 School-Level Analytics

- Performance across school sections.
- Performance by class and subject.
- Comparison between academic terms and examination sets.
- Academic trends across the school.
- Identification of classes requiring attention.
- Identification of subjects with consistently poor performance.
- Academic performance reports for management.

The system should support comparisons between Primary and Secondary education where applicable.

9. Early Identification of Students Falling Behind

One of the most important features is the ability to identify students who need academic support before final examinations. The system should use configurable performance thresholds.

Example Range	Interpretation
Below 40%	Critical academic concern
40–49%	Requires improvement
50–69%	Developing performance

Example Range	Interpretation
70–84%	Good performance
85% and above	Excellent performance

These thresholds are examples only. They should be configurable by school management and should not be treated as universal NECTA grading rules.

For each student requiring attention, the system should indicate current performance, previous performance, subjects with difficulties, topics requiring improvement, questions frequently answered incorrectly, and recommended academic intervention.

The purpose is not merely to label students as weak, but to help teachers understand what support each student needs.

10. Academic Intervention and Improvement

The platform should support academic intervention workflows. When a student is identified as falling behind, the teacher or academic administrator should be able to:

- Review the student's performance.
- Identify subjects and topics requiring improvement.
- Record an intervention plan.
- Assign remedial teaching or additional academic activities.
- Set an expected improvement target.
- Conduct a follow-up assessment.
- Compare performance before and after intervention.

Assessment → Analysis → Identification of Learning Gaps → Intervention → Follow-up Assessment → Progress Evaluation

The system should help the school determine whether academic interventions are producing measurable improvements.

11. User Roles and Access Control

School Administrator

- Manage school information, users, permissions, academic years, terms, classes, and subjects.
- Configure academic performance thresholds.
- Access school-wide analytics.

Academic Master / Academic Coordinator

- Monitor examination performance.
- Review academic reports.
- Identify struggling classes and subjects.
- Monitor assessment activities.
- Coordinate academic interventions.

Teacher

- Manage assigned classes and subjects.
- Create examinations and add questions.
- Record examination marks.
- View student performance.
- Review topic and question analytics.
- Identify students requiring support.
- Record academic interventions.

Student — Potential Future Role

- View examination results.
- Track academic progress.
- Review areas requiring improvement.

Parent / Guardian — Potential Future Role

- View academic performance.
- Monitor progress across examination sets.
- Receive academic progress reports.
- Understand areas where their children need support.

12. Reports and Visualization

The system should provide clear academic reports and visualizations, including:

- Individual Student Academic Report.
- Class Performance Report.
- Subject Performance Report.
- Topic Analysis Report.
- Question Analysis Report.
- Examination Results Report.
- Academic Progress Report.
- Students Requiring Intervention Report.
- Termly Academic Performance Report.
- School Academic Performance Report.

Reports should be exportable in formats such as PDF and Excel. The system should support charts and tables showing academic progress across examination sets.

13. Web Application and Future Mobile Application

Initial Development: Web Application

The web application will serve as the primary platform for teachers, school administrators, academic coordinators, and school management. It should provide examination management, results entry, analytics, reporting, and academic intervention management.

Future Development: Mobile Application

The mobile application may be introduced later, primarily for academic information access and analytics. Potential features include student performance summaries, academic progress charts, examination results, subject performance, topic-level learning gaps, academic notifications, and parent access to reports.

The mobile application will initially focus on analytics rather than full examination administration.

14. Proposed Technology Stack

- Frontend: React.js
- Backend: Node.js
- Database: PostgreSQL
- API: REST API
- Authentication: Secure role-based authentication
- Reporting: PDF and Excel generation
- Deployment: Cloud-based infrastructure

The architecture should be modular, secure, scalable, and maintainable. The database should support multiple school sections, academic years, classes, subjects, examination sets, questions, students, and performance records.

15. Important Design Principles

- Academic performance should be based on actual examination data.
- No simulated examination results or academic records should be used in production.
- Teachers should control examination creation and question selection.
- Analytics should be based on recorded marks and responses.
- The system should distinguish overall examination performance from topic-level performance.
- Academic thresholds should be configurable.
- Historical examination results should remain available for comparison.
- Student records should be protected through appropriate access control.
- Reports should be understandable to teachers, administrators, and parents.
- The platform should prioritize actionable academic insights rather than excessive dashboards or unnecessary statistics.

16. Expected Outcomes

- Improve academic performance through data-driven teaching.
- Identify struggling students earlier.
- Understand common learning difficulties.
- Improve examination preparation.
- Monitor academic progress across the school.
- Support teachers with evidence-based decisions.
- Evaluate the effectiveness of remedial teaching.
- Improve communication between teachers, management, and parents.

- Make academic planning more effective.
- Prepare students more effectively for NECTA examinations.

17. Long-Term Vision

The long-term vision is to transform Musabe Schools into a data-driven educational institution where academic decisions are based on detailed and reliable student performance information.

Rather than waiting for final examination results to identify academic problems, the school should be able to continuously monitor learning progress and take corrective action throughout the academic year.

The platform should eventually become a comprehensive academic intelligence system capable of analyzing years of examination data, identifying long-term performance trends, and helping the school continuously improve its teaching and learning outcomes.

The ultimate goal is not simply to record students' examination marks, but to understand their learning progress, identify academic weaknesses, and provide the right support at the right time.